CHM1001 L1st

velocity (n.)速率

nature (n.)本质

Planck's formula:

$$P(\nu,T)=\frac{8\pi h}{c^3}\cdot\frac{\nu^3}{e^{h\nu/kT}-1}$$

The Hydrogen Spectrum

$$\frac{1}{\lambda}=(R_H)\left(\frac{1}{n_1^2}-\frac{1}{n_2^2}\right)$$

The Bohr Model

$$\Delta E=-hcR_H\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

(final, initial) energy level

└the energy absorbed/emitted from...

e.g. $n=\infty$, $n=2$, $\Delta E>0$, $\Delta E<0$, $n=1$

quanta (p.l.)量子

└(singular: quantum)

photoelectric (adj.)光电的

spectrum

photon

excited states 激发态

integer

corresponding

ground states 基态

radiate

nucleus (n.)原子核

discrete

correspond

transition (n.)转变；跃迁

proportional (adj.)成比例的

solvent (n.)溶剂

tumor (n.)肿瘤

paradigm (n.)范例

reductionism (n.) 简化论
interpret (v.) 诠释
respectively (adj.) 分别地
infrared (adj.) 红外线的
region (n.) 地区，区域，领域
the wave nature of light 光的波动性
the nature of energy 能量的__
(半特质(味觉?)

(?) 2 W2P27

peak 波峰 (n.)
trough (n.) 波谷
(T=) periodic (n.) 周期

$c = \lambda \nu$

quantum (n.) 量子

$E = nh\nu$
↓
number of electrons

electromagnetic spectrum

photoelectric effect
photon

work function 逸出功 (W路：功函数)
dual wave particle nature 波粒二象性

spectrum (n.) 光谱；连续光谱；线状光谱
monochromatic (adj.) 单色的
polychromatic (adj.) 多色的
continuous spectrum 连续光谱
line spectrum 线状光谱

Bohr's model

the energies corresponding to the allowed orbits for the electron in the hydrogen atom

$E = (-hcR_H)\left(\frac{1}{n^2}\right)$
$= (-2.18 \times 10^{-18}\ J)\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

ground state 基态
excited state 激发态

the energy change

momentum 动量：$p = mv$

matter waves 物质波：$\lambda = \frac{h}{mv}$

uncertainty principle 测不准原理：

$$\underset{\downarrow}{\Delta x} \cdot \underset{\downarrow}{\Delta(mv)} \geq \frac{h}{4\pi}$$

位移　动量

wave function 波函数：
a series of mathematical

probability density
(electron density) (ψ^2)
(ψ (psi))

orbital

the principle quantum number (n) ⇒能层
主量子数
energy level, $n = 1, 2, 3, \cdots$

the angular momentum ⇒能级
quantum number (l)
角动量量子数
shape (of orbital)

the magnetic quantum number (m_l) ⇒轨道角向
磁量子数

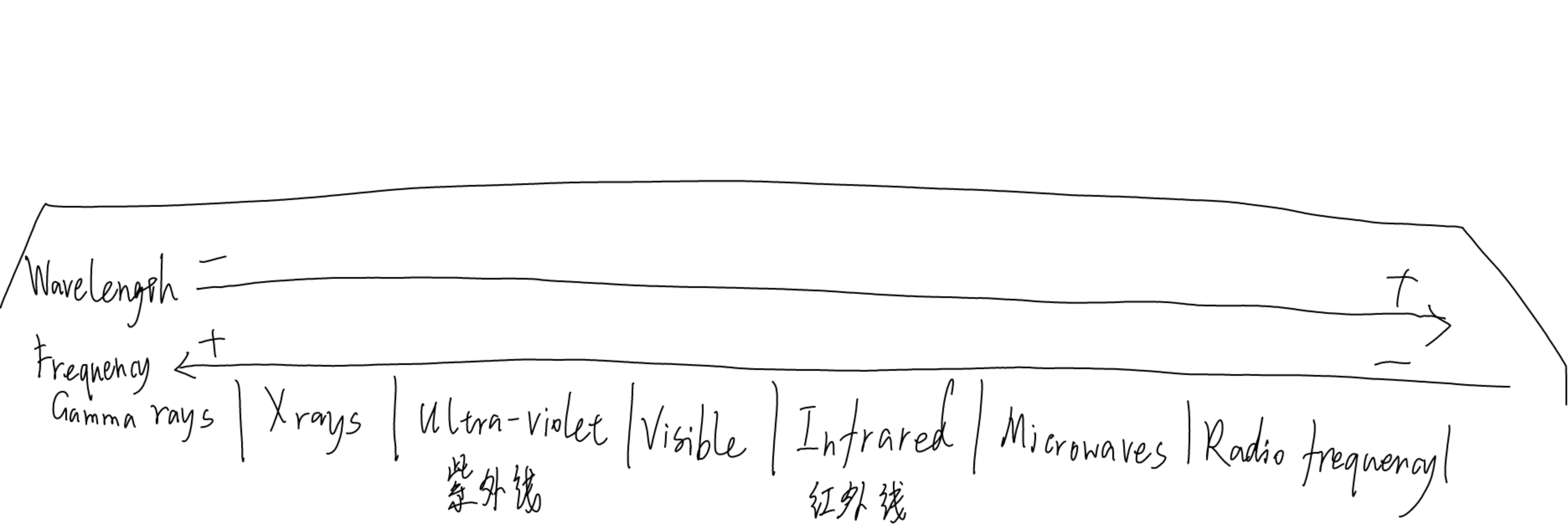
electron spin 电子自旋

the spin magnetic quantum number (m_s)
自旋磁量子数
$-\frac{1}{2}$ or $+\frac{1}{2}$ <

electron configuration 电子排布

orbital diagram 轨道图
(?电子排布图)

the Pauli exclusion principle
泡利不相容原理
Hund's rule


Wavelength
Frequency
Gamma rays
X rays
Ultra-violet
紫外线
Visible
Infrared
红外线
Microwaves
Radio frequency

W2 PPT - P4-2走位

$Z_{eff} = Z - S$

Ionization Energy 电离能

ground state 基态

gaseous (adj.) 气体的

successive (adj.) 连续的

valence electron 价电子

group 族

period 周期

nucleus 原子核

effective nuclear charge 有效核电荷

Electron Affinity 电子亲和力

exothermic (adj.) 放热的

[Metals]

luster (n.) 光泽

malleable (adj.) 有延展性的

ductile (adj.) 易延展的

conductivity

electrode

array

electrophysiology

[Nonmetals]

brittle

electronegativity

anion

readily

(Group) 1A The Alkali Metals 碱金属

2A The Alkaline Earth Metals 碱土金属

6A The Oxygen Group 氧族

7A The Halogens 卤素

8A The Noble Gases 稀有气体

anion (n.) 阴离子

Born-Haber Cycle ☆

9.19 PPT

Covalent 共价的

metallic 金属的

Lewis Symbols ①(?电子结构式?)

denote

valence electron 价电子

the octet rule

Ionic formation

①(Na· + ·Cl: → Na^+ + $[:\ddot{Cl}:]^-$)

isoelectronic (adj.) 等电子的

Born-Haber Cycle

② Energetics of Ionic Bonding

未x1P

Lattice Energy 晶格能

↳ P_1

$E_{el} = k\frac{Q_1 Q_2}{d}$

Coulomb's Law

Covalent Bonding

共价键

electrostatic interactions

attraction

repulsion

Lewis Structure

H_2: H-H

Cl_2: :Cl-Cl:

Electrons on
Lewis Structure

Lone pairs 孤电子对

Bonding pairs 成键电子对
↓
represented by 2dots/1line

Multiple Bonds
single bonds
double
triple

Polar Covalent Bonds
极性共价键
{ δ+ δ-
H—F
或
+ →
H—F

flourine (n.) 氟
the flourine end of
the molecule
分子的氟端
electron denisty
电子密度

electronegativity
电负性
↓
the more polar is the bond

Dipole Moment 偶极矩
↳符号: μ

Dipole
$\mu = Qr$

☆ P17 ⓘ
& P18

Lewis structure
—定义: P24
Formal charge 形式电荷
= valence charge
$-\frac{1}{2}$(bonding electrons)
- all nonbonding electrons

electronegativity & formal charge:
↳ N>S => ✓ $[\ddot{N}=C=\ddot{S}]^-$ (-1, 0, 0)
× $[:N\equiv C-\ddot{S}:]^-$ (0, 0, -1)

resonance

resonance structure 共振結构

ozone (n.) 臭氧

benzene

depict (v.) 描述

hexagon (n.) 六边形

localized electrons

delocalized electrons

Octet Rule

└ ⑦ P29

phosphate

└ More than eight electrons:
can use d-orbitals to
make more than 4 bonds

└ covalent bond strength

bond enthalpy

average bond enthalpies

endothermic process

P36~39 来

TUT 02

W3 (1) PPT

resonance (?)

multiple structure

resonance structure

depicted

localized electron

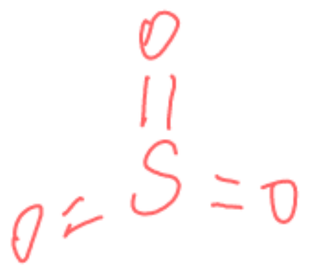
delocalized electron

W3 (2) PPT

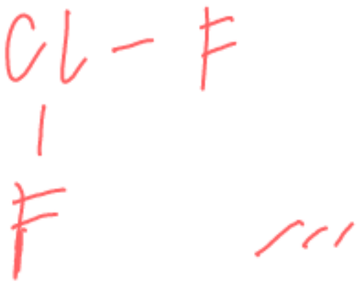
e.g.

AB_2 linear CO_2 O=C=O

AB_2 bent SO_2 

AB_3 trigonal planar SO_3 

AB_3 trigonal pyramidal NF_3 

AB_3 T-shaped ClF_3 

VSEPR: Valence-Shell Electron-Pair Repulsion

Electron Domains (?☆!)

└(P4)

↳ AB_2 linear

AB_3 trigonal planar

AB_4 tetrahedral

AB_5 trigonal bipyramidal

AB_6 octahedral

analogy (n.) 类比

repulsion

Electron-Domain Geometries (☆(P8~))

e.g.



linear

Molecular Geometries

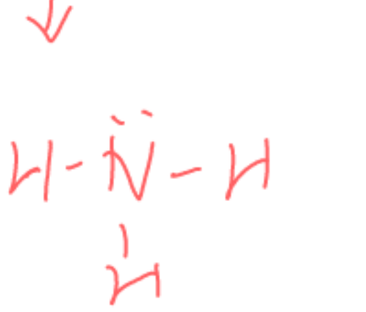
e.g.

NH_3

↓

Lewis Structure



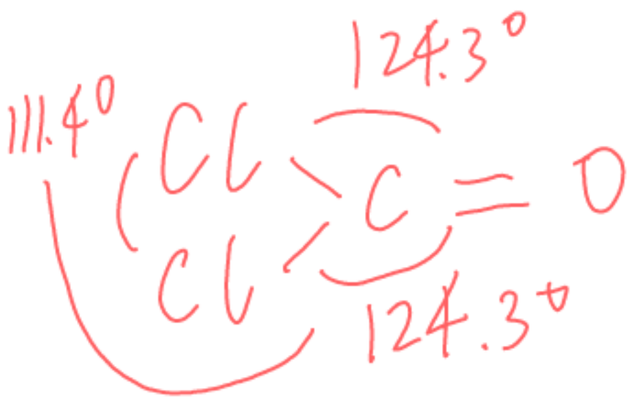


methane (n.) 甲烷

nonbonding pairs

bond angle /(s)

Multiple Bonds & Bond Angles

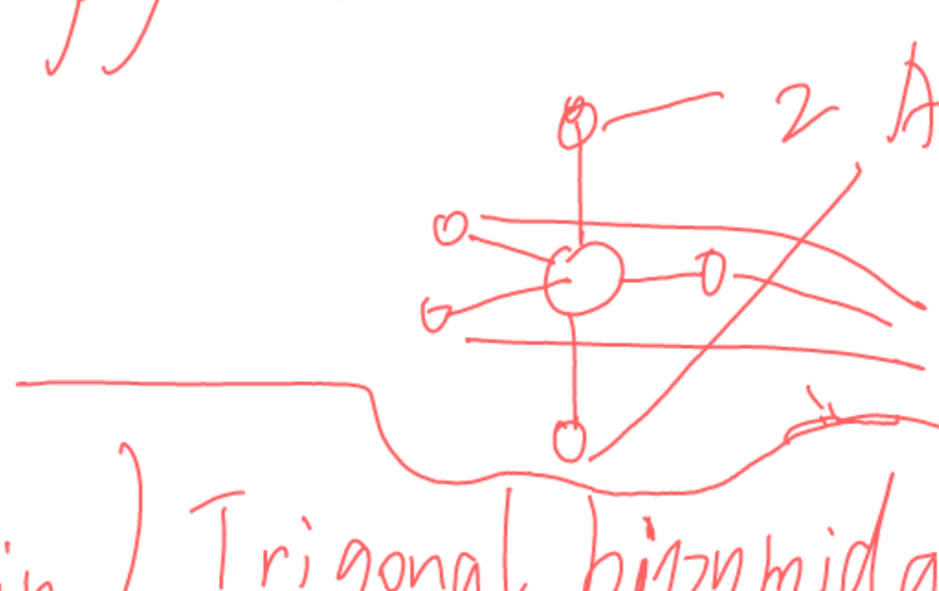


< *双、三键比单键

有更大 electron domains

exert

Trignal Bipymidal Electron Domain



Axial

Equatorial

Molecular Domain

Trigonal bipymidal	
Seesaw	SF_4
T-shaped	ClF_3
Linear	XeF_2

PCl_5 ...

seesaw ⑦

=(n.)跷跷板；拉锯局面

☆ Octahedral Electron Domain

↓

molecular geometries

{ x3 (P16)

Shapes of Larger Molecules?

★P18 polarity of molecules

electron density 电子密度

Valence-Bond Theory

2 atoms — occupy the same space



"overlap" of orbitals

(n.)重叠部分

The sharing of space ⇄ covalent bond

between

2 electrons — of opposite spin

Overlap & Bonding

electrons ——close—— nuclei (p.l.)原子核

↓

reach balance between

the like charge repulsions

⊗同极性电荷 (n.)斥力

&

the electron-nucleus attraction

VSEPR & Hybrid Orbitals

↓ 杂化轨道

(adj.)混合的(杂交的)

Hybrid Orbitals

mixing atomic orbitals

↓creat

new orbitals of equal energy

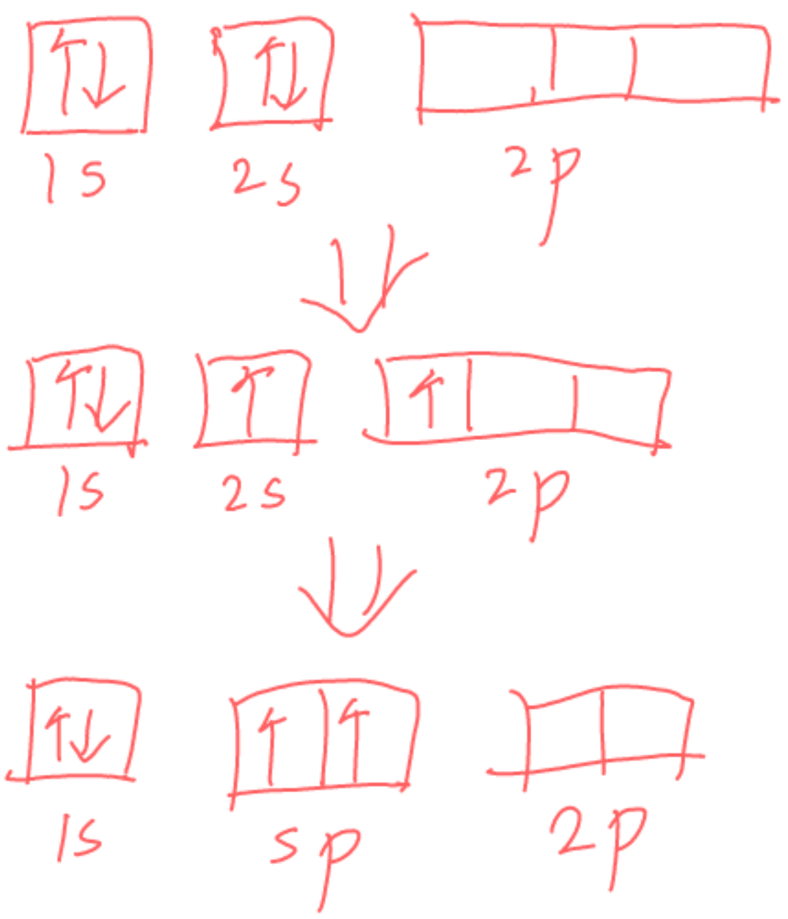
↓

degenerate orbitals

简并轨道

e.g. Be

-sp hybridization



sp Orbitals (p_{25})

lobe = (n.) 叶

s × 1 ⊙ + p × 1 ∞

↓ hybridize



sp × 2

one of the lobes:
larger, more rounded,
as is the s orbital

(sp hybrid orbitals superimposed ☆)
(adj.) 叠加的；重叠的
(large lobe only)

Position of sp Orbitals (p_{26})

e.g. BeF_2



sp^2

s × 1 ⊙ + p × 2 { ∞ (? $m_l \neq$ ☆)

↓ hybridize

sp^2 × 3 ⟶ show together



(trigonal planar)

(sp^3 同理)

× 4

↓



Hypervalent Molecules (P29)

(adj.)高价的

Types of Bonds (P32)

① "side-ways" overlap

- Sigma (σ) Bond
- p_i (π) Bond

σ Bond:
- head-to-head overlap
- cylindrical symmetry (n.)对称 of electron density about the internuclear axis

π Bond:
- side-to-side overlap
- ele… 未！P33

未！☆ P34

localized electrons
定域电子
└bonding electrons (σ or π)
that are specifically shared
between 2 atoms

来一P48完！
须看

delocalized electrons
不定域电子 (P35)
├aren't localized electrons
└shared by multiple atoms

Molecular Orbital (MO) Theory ?
Wave properties → 描述分子中电子的能量
characteristics X与 (P37，看！)

Contour diagram 等高图
contour = (n.) 等高线；轮廓